MOTOROLA SOLUTIONS ISRAEL LTD.
APPLIED RESEARCH IN MISSION CRITICAL ARTIFICIAL INTELLIGENCE

28LABS

28Labs is the Chief Technology Office and innovation engine for Motorola Solutions. We are user-focused and outcome-driven, identifying problems and creating solutions based on research and engagement with our customers.
Considering our customers' unique "jobs to be done" is a foundational principle of 28Labs. Based on their needs, we create solutions by applying our experience in user experience and design, advanced wireless communications and solutions, cloud & mobile application architecture, cybersecurity, user & human factors research, imaging, analytics, and artificial intelligence (AI).
Software and hardware technology and platform solutions developed by 28Labs form the basis of many products and solutions offered by Motorola Solutions.
28Labs was named in recognition of Motorola Solutions long history of innovation and creativity. Founded in 1928 by Paul and Joseph Galvin, the company's history is marked with technological leaps and groundbreaking inventions.
28Labs recognizes this legacy as we shape the future of Motorola Solutions by innovating across our portfolio and enabling new solutions for our broad customer base.

ISRAEL INNOVATION CENTER

Israel Innovation Center (IIC) is part of Motorola Solutions CTO organisation and global strategy for developing next generation advanced technology. IIC is a Research Organization for developing forward looking technologies and a Center of Excellence for AI, Deep learning, Machine Learning and Computer Vision.

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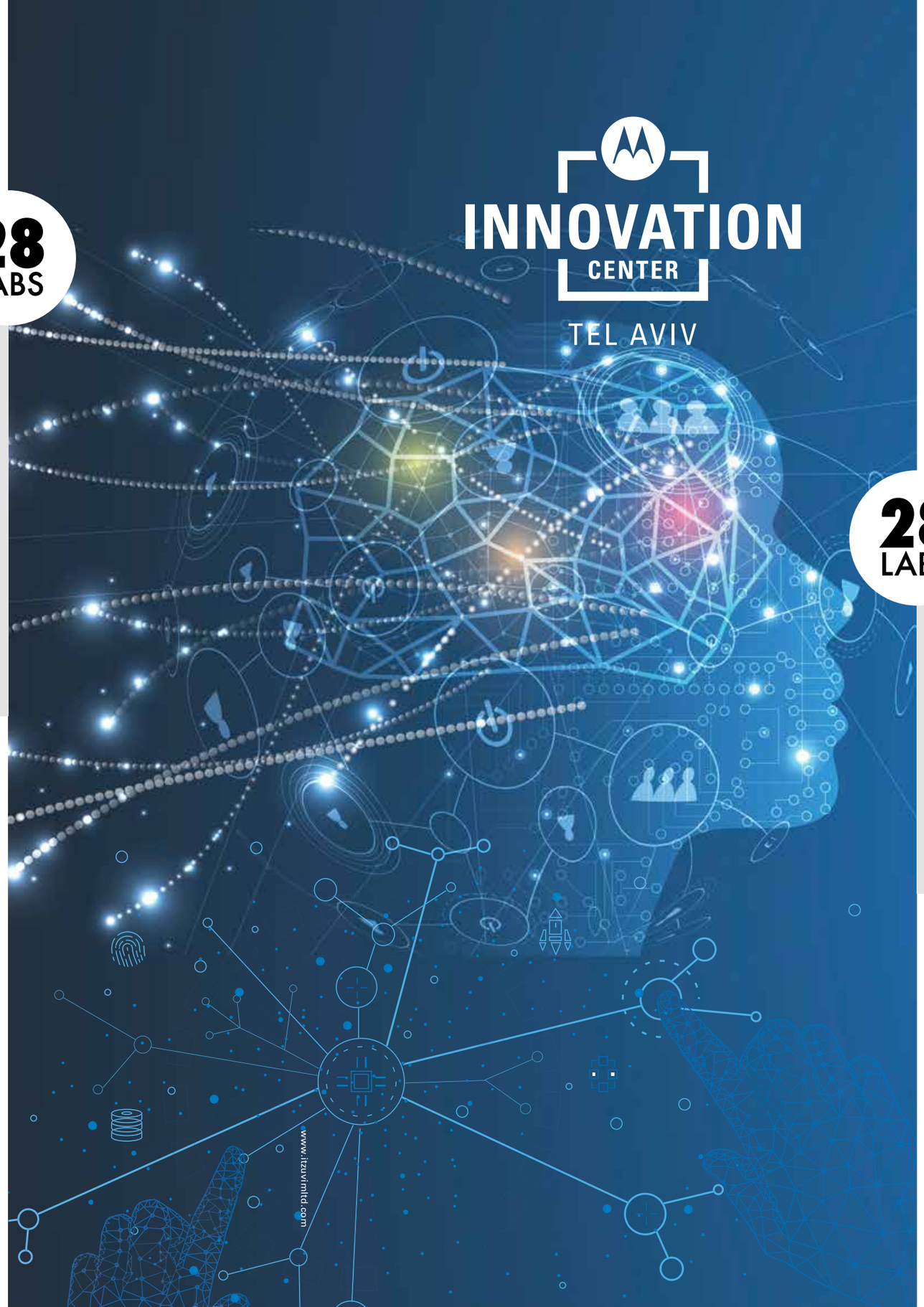

APPLIED RESEARCH IN MISSION CRITICAL ARTIFICIAL INTELLIGENCE

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ABOUT US

Israel Innovation Center (IIC) executes applied research projects for bringing artificial intelligence capabilities to a variety of current and future MSI products. IIC's engineers and researchers conduct advanced research that deals with various aspects of artificial intelligence applications including deep learning for vision, language and speech processing algorithms. The IIC taps into the Israeli ecosystem through collaborations with Israeli universities, startups and customers.

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Object Detection Before and After Haze Removal

PROJECT 1 Vision in bad weather for video analytics and visibility estimation

The opportunity: Mission Critical video analytics need to be operational 24/7 and, thus, have to cope with difficult weather conditions including fog, haze, smoke, rain and snow.

The objectives: Vision in bad weather is a joint project with the Tel-Aviv University. The project focuses on the development of image and video enhancement models for improving the accuracy of video analytics in bad weather conditions.

Top row: enhanced faces, Bottom row: original low-resolution faces

PROJECT 2 Enhancement of very low-resolution facial images for appearance search

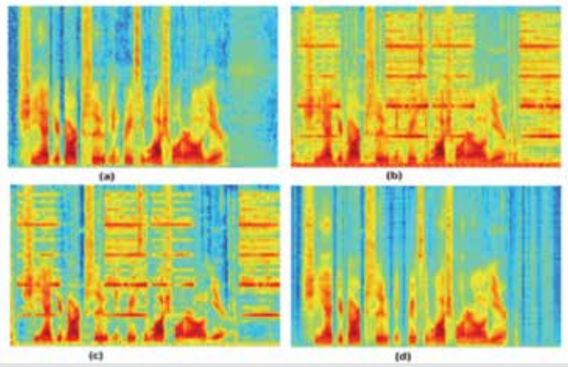
The opportunity: Video surveillance footage frequently includes distant, low quality or low resolution faces which need to be tracked and matched by the video analytics software.

The objectives: The project explores state-of-the-art deep learning super-resolution models for improving the accuracy of face re-identification algorithms.

PROJECT 3 Coded speech enhancement for improved automatic transcription of critical conversations

The opportunity: Coded compressed speech transmitted over radio or telephone links might lose signal components that are critical for automatic speech recognition (ASR). The ASR step is frequently the most basic processing step required for other AI-based information extraction algorithms applied to mission critical speech.

The objectives: This project explores special properties of coded speech for training a deep learning network for recovering lost content that is critical for a speech recognition engine.



Spectrograms (a) clean utterance (b) distorted speech (c) Log-MMSE (d) DNN Enhancement

PROJECT 4 Correction of speech recognition errors for the analysis of emergency call transcripts

The opportunity: Information extraction algorithms suffer from inaccuracies when applied to automatically generated emergency call transcripts. The inaccuracies are due to spontaneous, not always grammatically correct speech frequently used by people in high stress circumstances such as emergency call conversations.

The objectives: This project explores state-of-the-art deep learning sequence models for detecting and recovering speech transcription mistakes based on global and local conversational context.

SPEECH RECOGNITION OUTPUT	CORRECTED TRANSCRIPTS
I am living at three forty-one stoning heel road aa I am calling because the corpse is out earlier there was some guy up here he was coming to fight and stuff and my kids were getting ready to leave and they are trying to follow them back and forth.	I-am living (at>in) three forty one (stoning>stony) (heel>hill) road -aa -i (am>i'm) calling because the (corpse>cops) is out earlier (there>it) was -some -guy up here (he>there) was coming to fight -and (stuff>stafford) and -my (kids>martinez) (were>we're) getting ready to leave and they are trying to follow them back and forth.

Fragments of ASR-produced transcript and manually produced correct transcript

THOUGHT LEADERSHIP
IIC TOOK A LEADING PART IN SHAPING THE MSI AI ROADMAP THAT WAS PRESENTED IN THE CORPORATE CTO AI PAPER. "THE ARTIFICIAL INTELLIGENCE REVOLUTION AND IMPLICATIONS FOR MOTOROLA SOLUTIONS" (2017)

PROJECT 5 A software platform for robustness evaluation of speech, language and video processing algorithms

The opportunity: Mission Critical Artificial Intelligence algorithms should be robust and behave predictably when faced with difficult, distorted and noisy input frequently occurring in the context of public safety and other mission critical applications.

The objectives: This is an ongoing project for developing a unique platform for performing a robust evaluation of any AI algorithm. The platform (called Vulcan internally) provides services for augmentation and synthetic data generation based on state-of-the-art photorealistic and acoustic models. The platform stores and organize data and true labels, and allows to run robustness evaluation experiments that results in invaluable insights for algorithm development and improvement.

Screenshots of the Vulcan system showing evaluation data for vision (left) and language/speech (right) tasks